Mission Description Quality Catalog

Q1 – Atomicity: Measures the focus of each mission statement on a single, tangible idea or behavioral element that can be modeled.

Justification. Comparable to definitions found in the INCOSE Guide for Writing Requirements [1] and ISO/IEC/IEEE 29148:2018 [2], non-atomic mission statements are often the source of undefined behavior since interactions with other statements might be neglected. Also, a lack of atomicity hinders comprehension due to unseparated concerns. Atomicity measures the focus of each mission statement on a single self-contained portion of the robot's behavior.

4: *Atomic aspects* - Mission statements are self-contained and indivisible. Each mission statement expresses only a single aspect of the robot's behavior including preconditions.

3: *Moderately atomic* - Each mission statement focuses on a single key aspect but contains additional constraints or elements that can stand alone as a separate mission statement.

2: *Low atomicity* - Each mission statement consists of multiple aspects of behavior that are closely related via a simple relationship.

1: *Non atomic* - Each mission statement may contain multiple aspects of behavior that are only loosely related.

Q2 – Ambiguity: Mission statements should not have alternative meanings. They should be presented such that they can be interpreted in one way only. Unclear and vague terms like "fast" and "secure" must be avoided. Only explicit references and consistent node names must be used to avoid misconceptions in implementation details.

Justification. Comparable to general requirements (cf. ISO 29148, INCOSE Guide for Writing Requirements) [2, 1], mission statements that can be subjective may lead to inaccurate inferences. We observed such in some samples of reverse engineered mission descriptions, thus, its inclusion.

4: *No ambiguity* - The mission statement is explicitly defined, measurable and leaves no room for multiple interpretations.

3: *Low ambiguity* - The mission statement is mostly clear but has minor elements of misinterpretation.

2: *Moderate ambiguity* - The mission statement contains multiple possible interpretations, making it implementation and verification unclear.

1: *High ambiguity* - The mission statement is extremely vague making it nearly impossible to implement or interpret correctly.

Q3 – Completeness: A mission description must be capable of standing alone without any missing conditions or assumptions. The level of detail required for completeness may vary based on end-user preferences.

Justification. Completeness is an essential quality of both general (cf. ISO 29148 [2], ISO/IEC 25010 [3], IEEE 1061-1992 [4] and INCOSE Guide for Writing Requirements [1]) and mission descriptions. In our dataset analysis, we observed numerous inconsistencies that led to some incompleteness challenges.

4: *Explicit completeness* - The mission description explicitly contains all relevant details for understanding or implementing the described mission, without having to infer aspects from secondary sources or past practices.

3: *Implicit completeness* - The mission description contains all relevant details, but some are implicit. However, these implicit details are likely to be interpreted consistently across end users.

2: *Minor incompleteness* - The mission description outlines the overall mission comprehensively but omits certain details that should be specified either explicitly or implicitly.

1: *Incomplete* - The mission description omits major aspects of the mission.

Q4 – Understandability: This quality reflects how well target users can understand and accurately interpret a mission description, including whether technical users can reconstruct it as a behavior tree and non-technical users can grasp the robot's behavior within the mission context.

Justification. This aligns with the usability of general requirements found in ISO/IEC/IEEE 15288:2015 [5], IEEE 1061-1992 [4], ISO/IEC/IEEE 12207:2017 [6] and ISO/IEC 25010 [3]. The intuitiveness of a mission description can be assessed through usability, reflecting how effectively it conveys mission goals with minimal cognitive effort.

4: *Unrestrained understandability* - The mission description allows the end-user to infer all details necessary for a complete understanding of the mission.

3: *Understandable* - The mission description allows the end-user, depending on the scenario, to understand or model the general execution of the mission.

2: *Limited understandability* - The mission description conveys the overall mission but lacks sufficient detail for implementation.

1: *Not understandable* - The mission description does not enable end users to effectively operationalize it, for example by realizing it as a behavior tree.

Q5 – Consistency: This ensures that the mission statements of a mission description are non-contradictory, i.e., they do not contradict each other or the mission goals.

Justification. Consistency, as one of the most intensely researched qualities [7] supports correctness and understandability. It is thoroughly discussed in many standard requirements engineering documentation such as ISO/IEC/IEEE 29148 [2] and the INCOSE Guide for Writing Requirements [1].

4: *No inconsistency* - The mission description is not contradictory.

3: *Low inconsistency* - The mission description is mostly non-contradictory but has minor elements of contradiction.

2: *Moderate inconsistency* - The mission description contains multiple contradictions, causing incompatible behavior or operational decisions.

1: *High inconsistency* - The mission description is extremely inconsistent or contradictory, making it nearly impossible to implement or execute correctly.

References

[1] INCOSE, *Guide for Writing Requirements*, Version 4.

[2] ISO/IEC/IEEE 29148:2018, *Systems and software engineering – Life cycle processes – Requirements engineering*.

[3] ISO/IEC 25010, *Systems and software Quality Requirements and Evaluation (SQuaRE) – Product quality model*.

[4] IEEE 1061-1992, *Standard for a Software Quality Metrics Methodology*.

[5] ISO/IEC/IEEE 15288:2015, *Systems and software engineering – System life cycle processes*.

[6] ISO/IEC/IEEE 12207:2017, *Systems and software engineering – Software life cycle processes*.

[7] Montgomery et al. (citation key: Montgomery0076602100367).